Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Driven Multi-Omics Platform for Early Prediction of Congenital Autoimmune Diseases through Maternal–Infant Data Analysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio・Medical		Knowledge Service	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes /Bio-Equipment	Bio-electronics / Bio-Information	R&D / Engineering Services	Testing / Inspection / Analysis Techniques
Concept of Proposed Technology	The proposed technology is an **AI-driven multi-omics integration platform** designed to analyze maternal–infant biological data for the early prediction of congenital autoimmune diseases. The platform integrates **genomic data (DNA sequencing including HLA and immune-related variants), transcriptomic data (RNA sequencing), and microbiome profiles** obtained from maternal–neonatal dyads. By synthesizing transgenerational omics data, the platform models how **genetic susceptibility and early microbial colonization interact to shape neonatal immune development**. The system produces a unified analytical output, referred to as an **Integrated Immune Profile Score**, which quantitatively assesses autoimmune disease risk during early life. The technology shifts conventional healthcare approaches from post-onset diagnosis to **proactive, data-driven immune risk assessment**, enabling early monitoring and potential intervention during critical periods of immune system maturation.			

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Rationale for Joint R&D Project	This project requires international joint research and development due to the **scientific, clinical, and regulatory complexity** of congenital autoimmune disease prediction. First, while the Korean SME partner possesses strong capabilities in **genomic and microbiome analysis**, accurate interpretation of immune activity necessitates **high-level transcriptomic (RNA-seq) profiling**. Joint collaboration enables access to advanced RNA-sequencing expertise and novel isoform analysis essential for understanding functional immune responses. Second, **medical validation of AI-driven logic** is critical to prevent over-interpretation of biological data. Independent clinical verification ensures that the Integrated Immune Profile Score is conservative, reliable, and suitable for future clinical application. Third, **maternal–infant clinical datasets** are subject to strict ethical and regulatory constraints. Joint research allows access to established mother–infant cohorts and longitudinal biological samples, ensuring high-quality, transgenerational multi-omics data necessary for robust AI model training and validation.
Key Words	1. Maternal-Infant Vertical Transmission 2. Autoimmune Disease Prediction 3. Multi-omics (DNA/RNA/Microbiome) 4. RNA-seq Immune Profiling 5. AI-Driven Precision Medicine

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)

R&D Objectives

Overall Objective

To develop a **globally applicable AI-based multi-omics platform** capable of predicting congenital autoimmune disease risk by integrating maternal–infant genetic, transcriptomic, and microbiome data across diverse populations.

Specific Objectives

- Establish a **comprehensive transgenerational multi-omics database** incorporating maternal and neonatal DNA-seq, RNA-seq, and microbiome data
- Identify **genetic–microbial interactions** associated with autoimmune disease susceptibility
- Validate immune regulatory mechanisms through **transcriptomic analysis**
- Develop and optimize an **Integrated Immune Profile Score** that quantitatively assesses neonatal autoimmune risk
- Minimize population bias by accounting for **ethnic variability in biomarkers and genetic variants**

R&D Target Specifications

The project targets performance indicators related to **quantitative accuracy, sequencing sensitivity, analysis turnaround time, entity recognition accuracy, relationship extraction accuracy**, and the establishment of validated methodologies for immune risk prediction.

Key evaluation categories include:

- Correlation accuracy between sequencing methods
- Species-level microbiome detection sensitivity
- AI-based entity recognition and relationship extraction performance
- Methodological establishment for predictive immune scoring

NO.	EVALUATION ITEM (MAJOR PERFORMANCE)	UNIT	WEIGHT (%)	WORLD'S BEST
1	**Quantitative Accuracy (Amplicon NGS vs qPCR Correlation)**	R^2	15	0.872[1]
2	**Species-level Detection Sensitivity**	%	15	97
3	**Analysis Turnaround Time**	Hour	10	24
4	**Microbiome Entity Recognition (NER) Accuracy**	F1 Score	10	95.9[2]
5	**Immune Disease Entity Recognition (NER) Accuracy**	F1 Score	10	93.6[3]
6	**Immune Gene Entity Recognition (NER) Accuracy**	F1 Score	10	94.7[4]

	7	**Microbiome-Disease Relation Extraction (RE) Accuracy**	F1 Score	10	71.3[5]
	8	**Disease-Gene Relation Extraction (RE) Accuracy**	F1 Score	10	71.2[6]
	9	**Development of Measurement Methods and Target Setting for the Predictive Accuracy of the Integrated Immune Profile Score (IIPS)**	Qualitative	10	Methodology Establishment[7]

R&D Plan	**Year 1: Cohort Establishment and Data Infrastructure** • Establish joint governance and data-sharing frameworks • Finalize study design and ethical approvals for maternal–infant cohorts • Initiate biological sample collection and multi-omics sequencing • Develop standardized genomic, transcriptomic, and microbiome analysis pipelines • Construct a normalized database architecture for AI training **Year 2: Mechanistic Analysis and AI Model Development** • Expand multi-omics data generation to achieve statistical robustness • Curate clinically validated reference datasets • Conduct functional validation of genetic variants using RNA-seq • Integrate genetic susceptibility and microbiome data to identify immune activation mechanisms • Develop preliminary AI-based immune risk scoring logic and apply bias correction strategies **Year 3: System Integration and Global Validation** • Validate predictive performance across multi-ethnic datasets • Finalize the web-based multi-omics integration platform • Prepare technical documentation for future regulatory pathways • Establish joint intellectual property strategies for validated biomarkers

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will focus on **clinical validation, transcriptomic analysis, and global data reliability**, including: • Providing access to established maternal–infant cohorts • Leading advanced RNA-sequencing and transcriptomic analyses • Conducting cross-population mechanistic validation • Independently validating AI logic to ensure medical reliability and risk management
Expected R&R of Korean SME Partner	The Korean SME partner will focus on **AI development, system integration, and domestic data management**, including: • Managing domestic maternal–infant cohort data acquisition • Developing and optimizing AI-based immune risk scoring algorithms • Establishing genomic analysis pipelines for immune-related variants

	• Engineering the integrated web-based multi-omics platform • Benchmarking domestic data against global datasets

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Impact** • Establishment of a medically validated AI-based immune risk prediction framework • Deployment of a unified multi-omics integration system with high analytical reliability **Scientific and Clinical Impact** • Improved understanding of congenital autoimmune disease mechanisms • Identification of early immune activation markers linked to maternal–infant transmission **Economic and Industrial Impact** • Strengthened competitiveness of AI-driven bio-healthcare technologies • Foundation for future commercialization aligned with global regulatory standards **Social Impact** • Support for healthier early-life immune development • Reduction of long-term healthcare burden through early risk identification

6. References

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(2) John Snow Labs, "State-of-the-Art (SOTA) Natural Language Processing for Healthcare: Spark NLP," 2023.

(3) T. Kaur *et al.*, "3D CNN-based automated diagnosis of Parkinson’s disease using functional magnetic resonance imaging," *CNS Neuroscience & Therapeutics*, vol. 27, pp. 936-945, 2021.

(4) R. Parvez *et al.*, "A Hybrid Model for Named Entity Recognition in Medical Text," *Frontiers in Genetics*, 2022.

(5) J. Tu *et al.*, "Generative Relation Extraction for Microbiome-Disease Interactions," *Bioinformatics*, 2024.

(6) S. Duffield *et al.*, "geneDRAGNN: Gene Disease Relation Extraction using Graph Neural Networks," *Journal of Biomedical Informatics*, vol. 110, 2021.

(7) T. An et al., "Gut microbiota analysis reveals microbial signature for multi-autoimmune diseases based on machine learning model," *Frontiers in Microbiology*, vol. 16, 2025.